

PART 2 → Fitting (0,0.5), (2,3.5),
(1,1.5), (3,7.5)

INITIAL SUMS: $\sum x = 6$, $\sum x^2 = 14$, $\sum x^3 = 36$, $\sum x^4 = 98$

$\sum y = 13$, $\sum xy = 31$, $\sum x^2y = 83$

1. SETUP → LINE, $y = mx + b$

general formula: $MSE(m, b) = \frac{1}{n} \sum_{i=1}^n (y_i - (mx_i + b))^2$

$MSE(m, b) = \frac{1}{4} [(0.5 - b)^2 + (3.5 - (2m + b))^2 + (1.5 - (m + b))^2 + (7.5 - (3m + b))^2]$

2. SETUP → PARABOLA, $y = ax^2 + bx + c$

general formula: $MSE(a, b, c) = \frac{1}{n} \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c))^2$

$MSE(a, b, c) = \frac{1}{4} [(0.5 - c)^2 + (3.5 - (4a + 2b + c))^2 + (1.5 - (a + b + c))^2 + (7.5 - (9a + 3b + c))^2]$

① $y = mx + b$

analytical: set both partial derivatives to zero

$\frac{\partial MSE}{\partial m} = \frac{1}{n} \sum_{i=1}^n 2(y_i - (mx_i + b)) \times (-x_i)$

$= -\frac{2}{n} \sum_{i=1}^n (x_i y_i - mx_i^2 - bx_i)$

$= -\frac{2}{n} [\sum xy - m \sum x^2 - b \sum x]$

set to zero

$0 = -\frac{1}{2} [\sum xy - m \sum x^2 - b \sum x]$

$\sum xy = m \sum x^2 + b \sum x$

→ sub + solve system of equations

$14m + 6b = 31$

$6m + 4b = 13$

→ $b = \frac{13 - 6m}{4}$

$14m + \frac{6(13 - 6m)}{4} = 31$

⇒ $m = 2.3, b = -0.2, y = 2.3x - 0.2$

MSE calculation

(0, 0.5) → $y(0) = 2.3(0) - 0.2 = -0.2$

(1, 1.5) → $y(1) = 2.3(1) - 0.2 = 2.1$

(2, 3.5) → $y(2) = 2.3(2) - 0.2 = 4.4$

(3, 7.5) → $y(3) = 2.3(3) - 0.2 = 6.7$

residual

0.7

-0.4

-0.9

0.8

→ $MSE_{line} = \frac{0.49 + 0.36 + 0.81 + 0.64}{4} = 0.575$

numerical: alternating multivariate Newton-Raphson

→ fix b , solve $\frac{\partial MSE}{\partial m} = 0$ for m : $m = \frac{\sum xy - b \sum x}{\sum x^2} = \frac{31 - 6b}{14}$

→ fix m , solve $\frac{\partial MSE}{\partial b} = 0$ for b : $b = \frac{\sum y - m \sum x}{n} = \frac{13 - 6m}{4}$

ITERATION

	m	b
1	$m_1 = \frac{31 - 6(0)}{14} = 2.2143$	$b_1 = \frac{13 - 6(2.2143)}{4} = -0.07143$
2	$m_2 = \frac{31 - 6(-0.07143)}{14} = 2.2449$	$b_2 = \frac{13 - 6(2.2449)}{4} = -0.1173$
3	$m_3 = \frac{31 - 6(-0.1173)}{14} = 2.2646$	$b_3 = \frac{13 - 6(2.2646)}{4} = -0.1469$
4	$m_4 = \frac{31 - 6(-0.1469)}{14} = 2.2772$	$b_4 = \frac{13 - 6(2.2772)}{4} = -0.1658$
5	$m_5 = \frac{31 - 6(-0.1658)}{14} = 2.2854$	$b_5 = \frac{13 - 6(2.2854)}{4} = -0.1780$

→ sequence converging to 2.3 and -0.2, same as analytical ✓

$$\textcircled{2} y = ax^2 + bx + c$$

analytical : set partials to 0

$$MSE(a,b,c) = \frac{1}{n} \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c))^2$$

$$\begin{aligned} \frac{\partial MSE}{\partial a} &= \frac{2}{n} \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c)) \cdot (-x_i^2) \\ &= -\frac{2}{n} [\sum x_i^2 y - a \sum x_i^4 - b \sum x_i^3 - c \sum x_i^2] \end{aligned}$$

$$\Rightarrow a \sum x_i^4 + b \sum x_i^3 + c \sum x_i^2 = \sum x_i^2 y$$

$$98a + 36b + 14c = 83 \quad (1)$$

$$\frac{\partial MSE}{\partial b} = \frac{2}{n} \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c)) \cdot (-x_i)$$

$$\Rightarrow a \sum x_i^3 + b \sum x_i^2 + c \sum x_i = \sum x_i y$$

$$14a + 6b + 4c = 13 \quad (3)$$

$$\frac{\partial MSE}{\partial c} = \frac{2}{n} \sum_{i=1}^n (y_i - (ax_i^2 + bx_i + c)) \cdot (-1)$$

$$a \sum x_i^2 + b \sum x_i + c \sum 1 = \sum y_i$$

$$\Rightarrow 36a + 14b + 6c = 31 \quad (2)$$

solve system of equations, interim steps excluded:

$$a = 0.75, \quad b = 0.05 + c = 0.55$$

$$y = 0.75x^2 + 0.05x + 0.55$$

MSE calculation

$$(0, 0.5) \rightarrow y(0) = 0.55$$

residual

$$-0.05$$

$$(1, 1.5) \rightarrow y(1) = 1.35$$

$$-0.15$$

$$\rightarrow MSE_{\text{prob}} = \frac{2(0.05)^2 + 2(0.15)^2}{4}$$

$$(2, 3.5) \rightarrow y(2) = 3.45$$

$$-0.15$$

$$= 0.0125$$

$$(3, 7.5) \rightarrow y(3) = 7.45$$

$$0.05$$

numerical $\rightarrow$ cycle through parameters + iterate

$$a = \frac{83 - 36b - 14c}{98}$$

$$b = \frac{31 - 36a - 6c}{14}$$

$$c = \frac{13 - 14a - 6b}{4}$$

ITERATION

	a	b	c
1	$a_1 = \frac{83 - 36(0) - 14(0)}{98} = 0.8469$	$b_1 = \frac{31 - 36(0.8469) - 6(0)}{14} = 0.03644$	$c_1 = \frac{13 - 14(0.8469) - 6(0.03644)}{4} = 0.231$
2	$a_2 = \frac{83 - 36(0.0364) - 14(0.231)}{98} = 0.8005$	$b_2 = \frac{31 - 36(0.8005) - 6(0.231)}{14} = 0.0567$	$c_2 = \frac{13 - 14(0.8005) - 6(0.0567)}{4} = 0.363$
3	$a_3 = \frac{83 - 36(0.0567) - 14(0.363)}{98} = 0.774$	$b_3 = \frac{31 - 36(0.774) - 6(0.363)}{14} = 0.0678$	$c_3 = \frac{13 - 14(0.774) - 6(0.0678)}{4} = 0.438$
4	$a_4 = \frac{83 - 36(0.0678) - 14(0.438)}{98} = 0.759$	$b_4 = \frac{31 - 36(0.759) - 6(0.438)}{14} = 0.0736$	$c_4 = \frac{13 - 14(0.759) - 6(0.0736)}{4} = 0.482$

$\rightarrow$ sequence converging to analytical values $\checkmark$